

Gamification & Improving Hepatology Knowledge for Medical Students

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Theme 1: Gen Z Learning Needs

In order to teach current and upcoming medical students, it's important to recognize the learning needs of Gen Z learners (i.e., those born between 1997-2012) which represents a wide range of those currently aged 14-29. With a focus on adult learners, various review articles were identified, with the first being by Eckleberry-Hunt et al. (2018). Focusing on medical education and Generation Z learners, this review article highlighted specific educational needs of these learners including the need for customized learning or an 'a la carte' approach. Not only was this noted to be required from an educational methodology, but also specifically the need for customized feedback was highlighted. Generation Z learners, born in the digital age, are accustomed to getting information on demand and may lack the skills to critically evaluate information, especially when so much information can be readily found on the internet. Training is important for these learners in order to improve these skills to ensure that the information identified is factually correct. Finally, the benefits of a flipped classroom style of learning were mentioned especially as these learners may have reduced attention spans overall.

A second narrative review article, published by Shirazian et al. (2025) was also reviewed, with articles selected from Medline based on a search of published articles between 2000-2021. Similar to Eckleberry-Hunt et al. (2018), the benefits of a flipped classroom were highlighted as well as the importance of individualized or customized feedback to help with learner development. Storytelling, to include different perspectives, was also felt to be an effective strategy for these learners. Unlike prior generations, the traditional lecture platform is less

effective compared to other types of learning environments, especially as these learners are accustomed to multi-tasking and have decreased attention spans overall. Also, to improve learner motivation and attention, interactivity was highlighted as an important strategy.

Theme 2: Gamification & Medical Education

A narrative review article focused on gamification and generation Z medical education, published in 2024 by Sundareswaran et al. was identified for inclusion in this literature review. This article reviewed published articles related to various medical education games focusing on physiology. Continuing the themes identified by Eckleberry-Hunt et al. (2018), the benefits of new media (i.e., computer games, video games, VR environments, social networks) and medical education would be of particular benefit to Gen Z learners as they are digital natives. Similarly, the flipped classroom idea was highlighted as a potential effective means to motivate Gen Z learners. This review article also describes various software options for game development (such as Construct 3, Unity, Gamefroot, Gamestar Mechanic) with further details and sample gameplay images provided from GameStar Mechanic-developed games.

Another narrative review article, this time based on a PubMed search (search years not included) was published by Xu et al. (2023) focusing on a broader review of game-based learning in medical education – unlike the physiology focus described by Sundareswaran et al. (2024). 14 medical education games were evaluated, and importantly, mixed media were included including board games, offline games, and online games. What I particularly appreciated about this review was the increased focus on the weaknesses and limitations of game-based medical education. There are no published studies looking at long-term knowledge retention, and most importantly, no standardized evaluation metrics by which to measure the

impact of gamification in medical education or learning outcomes. Most publications regarding games focus on outcomes of learner satisfaction or degrees of engagement or pre-/post-testing but there are no studies looking at long-term knowledge gain. As well, most studies that are published are limited solely to single institutions.

Based on Xu et al. (2023), a sub-theme was identified – namely, the need for ways to test for validity of game-based medical education. A recent publication by Surapanemi (2026) reviewed the current evidence and future directions for gamification for assessment in medical education. Again, a narrative review article, though with a broad search scope including PubMed, Scopus, and Web of Science between 7/2015-7/2025 with 30 articles meeting inclusion criteria. 4 key domains were reviewed: Learner Engagement, Implementation Challenges, Validity & Reliability, Feedback & Motivation. In brief, there is very limited empirical evidence supporting the use of gamification in summative assessment as most published articles related to games were used for formative assessment. Also highlighted is the importance of learner feedback to effective game implementations, especially when allowing for immediate correction and progress tracking. Points or badges as motivational tactics or rewards can be helpful, but risk trivialization if they are simply integrated in a superficial manner. Narrative-based designs were noted to have a broader learner appeal; however, there is a strong need for “rigorous evaluation of validity, reliability, defensibility” if gamification is to be used for summative evaluation.

Theme 3: Hepatology Medical School Curricular Content – Status Quo

Though there were limited peer-reviewed publications in this area, Pan et al. (2022) conducted a very interesting prospective survey-based study at a single institution investigating the comfort level of recently graduated medical students on managing patients with chronic liver

disease as compared to other non-liver medical conditions. Over a 2-year period, incoming categorical internal medicine residents were surveyed using a 38-question survey based on Likert-style questions. As well, multiple choice questions taken from a board prep database (MKSAP) were used to objectively measure knowledge on liver-specific topics. Statistical analysis using Fisher's exact and Wilcoxon ranked sum tests was conducted for the 65 interns that completed the survey (100% response rate). In summary, incoming interns were found to be less comfortable managing liver diseases as compared to other medical conditions, with a minority of students reporting satisfaction with hepatology education during medical school

A sub-theme was identified based on this study, looking at effective means of improving knowledge gaps amongst medical residents with targeted lectures-based education noted to be of benefits (Darmadi et al., 2025); however, this does seem at odds with incoming Gen Z learners as noted earlier. In this quality-improvement project conducted at a single institution between January to February of 2025, metabolic-dysfunction associated steatotic liver disease (MASLD) was previously identified to be a weak area for medical residents, and a structured lecture on this topic was implemented into the standard curriculum. Learner assessment was conducted using paired multiple-choice questions on pre- and post-intervention surveys (analyzed with paired t-tests) and through 4-point Likert scale learner satisfaction questions. 111/125 residents completed both surveys, with strong improvements in learner confidence across 3 tested domains (obesity, anti-obesity medications, MASLD) with improvement in quiz scores post-lecture ($p < 0.001$). However, whether this remains an effective strategy for Gen Z learners given their previously noted learner needs is yet to be determined.

References:

1. Eckleberry-Hunt, J., Lick, D., & Hunt, R. (2018). Is Medical Education Ready for Generation Z?. *Journal of graduate medical education*, 10(4), 378–381.
<https://doi.org/10.4300/JGME-D-18-00466.1>
2. Sundareswaran, L., Krishnan, S., Sinha, A., Naveen, P., Mahanta, A., & Bhattacharjee, M. (2024). Making a serious game (gamification) for generation Z medical students to learn, teach, and assess medical Physiology. *Journal of education and health promotion*, 13, 212. https://doi.org/10.4103/jehp.jehp_1177_23
3. Shirazian, S., Vatanpour, M., & Guity, P. (2025). Generation Z: Learners' Profiles and Teaching Challenges. *Frontiers in dentistry*, 22, 40.
<https://doi.org/10.18502/fid.v22i40.19906>
4. Pan, A. Y., Zilberstein, N. F., Farnan, J. M., McConville, J. F., & Mikolajczyk, A. E. (2022). Recently Graduated Medical Students Lack Exposure to and Comfort with Chronic Liver Diseases. *Digestive diseases and sciences*, 67(6), 2081–2085.
<https://doi.org/10.1007/s10620-021-07182-0>
5. Xu, M., Luo, Y., Zhang, Y., Xia, R., Qian, H., & Zou, X. (2023). Game-based learning in medical education. *Frontiers in public health*, 11, 1113682.
<https://doi.org/10.3389/fpubh.2023.1113682>
6. Surapaneni, K. M. (2026). Gamification for Assessment in Medical Education: Exploring Current Evidence and Future Directions from Broader Educational Contexts. *Current Opinion in Physiology*, 100907.
7. Darmadi, M., Potluri, V., Thomas, R. C., Park, C. J., Oladunjoye, O. O., & Thomas, R. (2025). Assessing and Addressing Medical Residents' Knowledge Gaps in the Screening

and Treatment of Metabolic Dysfunction-Associated Steatotic Liver Disease, Obesity, and Metabolic Syndrome: A Quality Improvement Project. *Cureus*, 17(12).